

# BB84 Quantum Key Distribution

*Qiskit Aer Simulation — Protocol, Attack Model, and Security Analysis*

## 1. Protocol Description

BB84 is simulated end-to-end at the circuit level in Qiskit Aer. Alice generates a random bit string and a random basis string (Z / computational or X / Hadamard) and encodes each bit as a single-qubit state accordingly. Each qubit is then routed through an optional eavesdropping stage and an optional noisy channel before Bob measures it in an independently random basis.

Eve is modeled as an intercept-resend attacker: for each qubit she intercepts with probability  $\eta$ , she measures it in a randomly chosen basis and forwards the post-measurement state, unavoidably disturbing it whenever her basis disagrees with Alice's. Independently, each qubit may suffer depolarizing channel noise with probability  $p$ , applying a uniformly random Pauli fault (X, Y, or Z).

After measurement, Alice and Bob publicly compare bases and keep only the positions where they agree (sifting). A random 25% subset of the sifted bits is sacrificed to estimate the quantum bit error rate (QBER); if the sampled QBER meets or exceeds the 11.04% asymptotic Shor–Preskill security bound, the protocol aborts and no key is produced. Otherwise the remaining raw key is passed through multi-pass Cascade error reconciliation (leaking a tracked number of parity bits) and then compressed with a 2-universal Toeplitz hash for privacy amplification, yielding the final secret key.

### 1.1 Secret Key Rate

The fraction of the raw key retained after privacy amplification follows the asymptotic Shor–Preskill formula, which bounds the net information advantage Alice and Bob hold over Eve after accounting for reconciliation leakage:

$$R = \max(0, 1 - (1 + f) \cdot h_2(Q))$$

where  $Q$  is the estimated QBER,  $f = 1.16$  is the reconciliation inefficiency of the Cascade protocol relative to the Shannon limit, and  $h_2$  is the binary Shannon entropy.  $R$  is bounded to the range  $R \in [0, 1]$ :  $R = 1$  only in the noise-free, attack-free limit ( $Q = 0$ ), and  $R = 0$  once  $Q$  reaches the 11.04% security threshold — beyond that point privacy amplification cannot remove Eve's information faster than it destroys Alice and Bob's shared bits, so no secure key can be extracted.

### 1.2 Simulation Parameters

Each operating point ( $\eta$ ,  $p$ ) is evaluated over 20 independent Monte-Carlo trials to obtain stable mean  $\pm$  standard-deviation estimates rather than a single noisy sample:

| Parameter            | Value               | Meaning                                                   |
|----------------------|---------------------|-----------------------------------------------------------|
| n_qubits             | 1000 / trial        | Qubits sent by Alice per trial                            |
| sample_ratio         | 0.25                | Fraction of sifted bits sacrificed for QBER estimation    |
| N_TRIALS             | 20 / point          | Monte-Carlo repeats per ( $\eta$ , noise) operating point |
| f_leak               | 1.16                | Reconciliation inefficiency factor (Cascade overhead)     |
| QBER abort threshold | 11.04 %             | Asymptotic Shor–Preskill security bound                   |
| $\eta$ (eta)         | 0.0 – 1.0, step 0.1 | Probability Eve intercept-resends a given qubit           |

|                        |               |                                          |
|------------------------|---------------|------------------------------------------|
| Depolarizing noise $p$ | 0 %, 2 %, 5 % | Independent channel-noise profiles swept |
|------------------------|---------------|------------------------------------------|

## 2. Results

### 2.1 QBER vs. Attack Strength

Figure 1 sweeps Eve's intercept probability  $\eta$  from 0 to 1 at three channel-noise levels (0%, 2%, 5%). The simulated QBER rises linearly with  $\eta$ , closely tracking the theoretical prediction  $\text{QBER} \approx p/2 + (\eta/4)(1 - p)$  (dashed lines). Error bars ( $\pm 1$  std over 20 trials) are widest in the mid-range of  $\eta$ , where individual trials straddle the security threshold and sampling variance is largest; they shrink toward both ends, where outcomes are more deterministic.

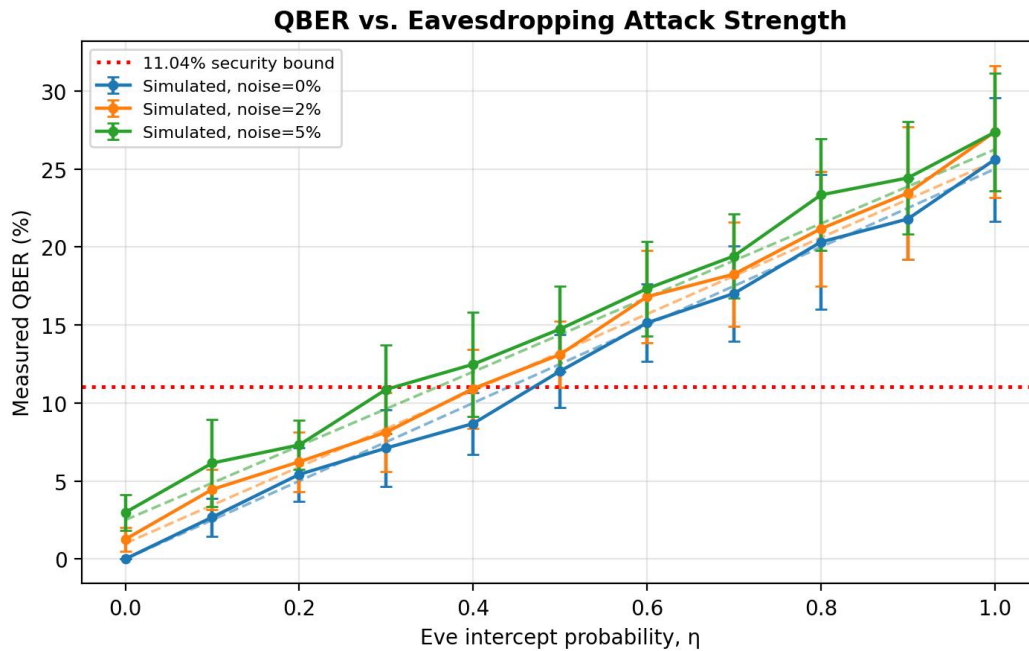


Figure 1. Measured QBER vs. Eve's intercept probability  $\eta$ , for three depolarizing-noise levels. The red dotted line marks the 11.04% Shor–Preskill security bound; simulated points (solid) track the closed-form theoretical prediction (dashed).

### 2.2 Secret Key Rate vs. Noise & Attack Strength

Figure 2 shows the resulting final secret key rate (bits of secure key per pulse sent) over the same sweep. The key rate falls sharply as  $\eta$  increases, reaching zero at  $\eta \approx 0.44$ – $0.5$  across all three noise profiles — consistent with the QBER crossing the 11.04% abort threshold in Figure 1. Higher intrinsic channel noise shifts the curve down and left: it erodes the key rate even at  $\eta = 0$  (from 0.373 bits/pulse at 0% noise to 0.221 bits/pulse at 5% noise) and causes the protocol to abort at a slightly lower  $\eta$ .

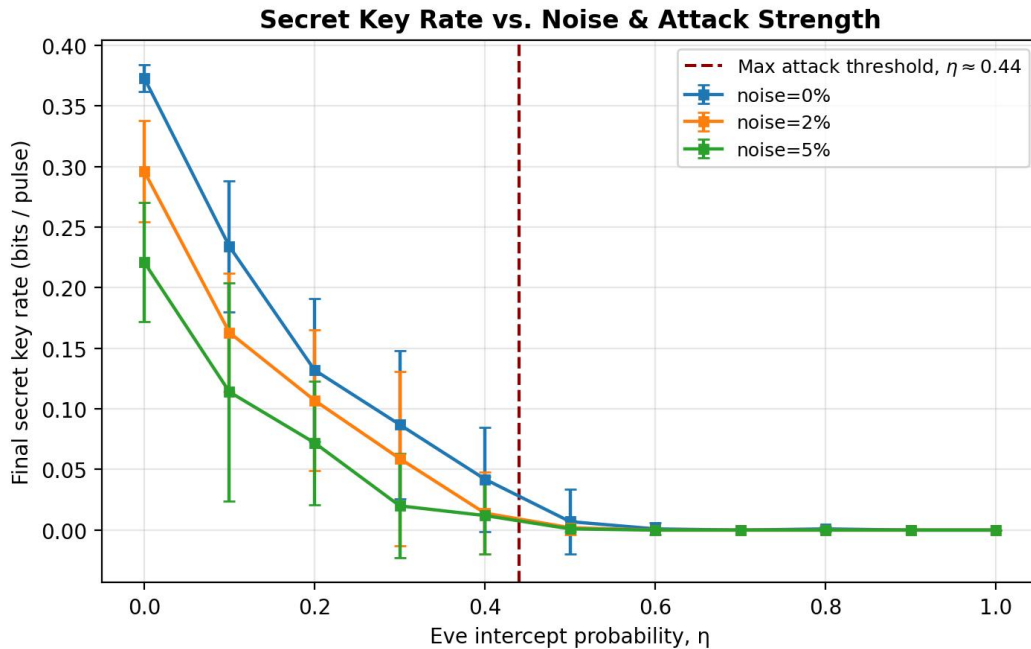


Figure 2. Final secret key rate vs.  $\eta$  for the same three noise profiles. The dashed vertical line marks the empirical point beyond which the key rate is statistically indistinguishable from zero.

### 3. Security Discussion

#### 3.1 What the threshold means

The 11.04% QBER cutoff is the point at which  $R = 1 - (1+f)h_2(Q)$  reaches zero for  $f = 1.16$ : past it, Cascade's reconciliation leakage plus the entropy Eve could hold exceeds the raw key size, so privacy amplification would need to compress the key to nothing. Aborting here is a security feature, not a failure — it guarantees that any key that is produced is provably secret in the asymptotic, collective-attack model.

#### 3.2 Distinguishing noise from eavesdropping

Because  $\eta$  and channel noise  $p$  both raise QBER, the protocol cannot directly tell whether elevated errors come from a noisy channel or an active attacker — only the aggregate error rate is observable. At the noise levels simulated here (0–5%), this is not a practical problem: even 5% intrinsic depolarizing noise alone yields a baseline QBER of only ~2.5–3%, well inside the operational regime. A false abort from noise alone would require unrealistically high hardware noise (baseline QBER approaching 11% needs depolarizing noise beyond 15–20%), so in this simulation every abort near the threshold is attack-driven rather than a noise artifact.

#### 3.3 Assumptions and limitations

- Attack model: only intercept-resend is simulated. It is the simplest individual attack; the 11.04% bound itself is derived against the strictly stronger class of collective attacks, so the simulated aborts are conservative relative to what full coherent-attack analysis would require.
- Asymptotic key-rate formula:  $R = 1 - (1+f)h_2(Q)$  assumes infinitely long keys. At the simulated block size (~1000 qubits, ~250 sifted-and-sampled bits per trial), finite-key statistical fluctuations are real — which is exactly why the report averages 20 trials per point rather than trusting a single run.

- Idealized photon source: each pulse is treated as a genuine single photon. Real weak-coherent-pulse implementations are vulnerable to photon-number-splitting attacks not captured here; production systems mitigate this with decoy states.
- Reconciliation overhead is fixed:  $f = 1.16$  models Cascade's typical inefficiency but does not vary with block size or QBER, as it would in a full implementation.
- No channel loss or detector imperfections (dead time, dark counts, detector blinding) are modeled — only bit errors from noise and eavesdropping.
- The classical sifting / reconciliation channel is assumed to be authenticated; BB84 does not provide authentication itself and none is simulated here.

### 3.4 Practical takeaway

Across all three noise profiles, the protocol reliably distinguishes secure from compromised channels: the key rate collapses to zero within a narrow  $\eta \approx 0.44\text{--}0.5$  band that lines up with the QBER crossing 11.04%, and it does so whether or not additional channel noise is present. This is the behavior a correct BB84 implementation should exhibit — a smooth, monotonic tradeoff between attack strength and extractable key, with a hard, provable cutoff rather than a silent degradation in security.